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## Unit “ 3 : Session “ 5 : SLO “ 1 | Infix to Postfix Conversion “” Worked Example

### ■ I. Step-by-Step Trace: $A - (B / C + (D \% E * F) / G) * H$

- A -> operand | Output: A | Stack: -
- - -> push | Output: A | Stack: -
- ( -> push | Output: A | Stack: - (
- B -> operand | Output: AB | Stack: - (
- / -> push | Output: AB | Stack: - (/
- C -> operand | Output: ABC | Stack: - (/
- + -> pop /, push + | Output: ABC/ | Stack: - (+
- ( -> push | Output: ABC/ | Stack: - (+ (
- D -> operand | Output: ABC/D | Stack: - (+ (
- % -> push | Output: ABC/D | Stack: - (+ ( %
- E -> operand | Output: ABC/DE | Stack: - (+ ( %
- \* -> pop %, push \* | Output: ABC/DE% | Stack: - (+ ( \*
- F -> operand | Output: ABC/DE%F | Stack: - (+ ( \*
- ) -> pop till ( | Output: ABC/DE%F\* | Stack: - (+
- / -> pop +, push / | Output: ABC/DE%F\*+ | Stack: - (/
- G -> operand | Output: ABC/DE%F\*+G | Stack: - (/
- ) -> pop till ( | Output: ABC/DE%F\*+G/ | Stack: -
- \* -> push | Output: ABC/DE%F\*+G/ | Stack: - \*
- H -> operand | Output: ABC/DE%F\*+G/H | Stack: - \*
- END -> pop \*, pop - | Output: ABC/DE%F\*+G/H\*- | Stack: empty

### ■ II. Final Answer

- Postfix expression:  $A B C / D E \% F * G / + H * -$
- Compact form:  $ABC/DE\%F*+G/H*-$